Lab experiment

4.Implementation of java program using multithreading

Aim:

To write a Java program to implement multithreading by creating and executing multiple threads simultaneously.

Algorithm:

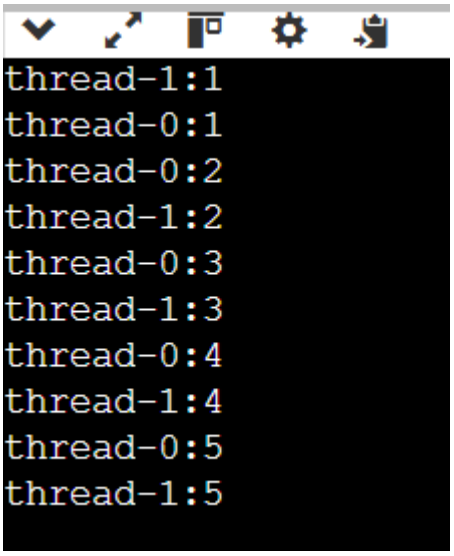
1.Start the program.
2.Create a class that extends the `Thread` class.
3.Override the `run()` method to define the task performed by each thread.
4.Create two thread objects.
5.Start both threads using the `start()` method.
6.Each thread executes its task concurrently and displays its thread name and count.
7.Wait for both threads to complete execution using `join()`.
8.Stop the program.

Coding:

```
class MyThread extends Thread {
    public void run() {
        for (int i = 1; i <= 5; i++) {
            System.out.println(getName() + ":" + i);

            try {
                Thread.sleep(100);
            } catch (InterruptedException e) {
                System.out.println(e);
            }
        }
    }
}

public class Multithreading {
    public static void main(String[] args)
            throws InterruptedException {

        MyThread t1 = new MyThread();
        MyThread t2 = new MyThread();

        t1.start();
        t2.start();
```

```
        t1.join();
        t2.join();
    }
}
```

Output:



Result:

Thus, a Java program implementing multithreading was executed successfully by creating and running two threads concurrently.